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Perceived Epidemic Impacts and Mental Symptom Trajectories in Adolescents Back to School After COVID-19 Restriction: A Longitudinal Latent Class Analysis

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A B S T R A C T

Purpose: This study aimed to assess the impacts of COVID-19 epidemic on various life aspects and identify the trajectories of common mental symptoms among adolescents back to school after COVID-19 restriction. Furthermore, potential predictors associated with those trajectories were investigated.
Methods: This longitudinal study, with five data collection points and a total follow-up of 68.4 days, was conducted among 1,393 junior high school students (mean age: 13.8 years; male, 53.3%) shortly after school reopened during the first COVID-19 outbreak in China. Questions on sociodemographics and perceived COVID-19 epidemic impacts were completed at the baseline while the Patient Health Questionnaire, Generalized Anxiety Disorder Scale, and Insomnia Severity Index were measured throughout the study for depression, anxiety, and insomnia symptoms, respectively. Trajectories of mental symptoms were classified by longitudinal latent class analysis, and the associated predictive factors were identified with multinomial regression modelling.
Results: Our study revealed high but steadily declining prevalence of depression, anxiety, and insomnia symptoms (p trend $< .001$). Five distinctive trajectories were identified for both depression and anxiety ("resistance," "low symptom," "recovery," "chronic dysfunction," and "delayed dysfunction") and three for insomnia ("resistance," "low symptom," and "chronic dysfunction"). Besides the significant association between the mental symptom trajectories and students' perceived COVID-19 impacts on study practice, family income, and family relationship, female gender, lower school grade, and higher body mass index were found to be predictive of high severity trajectories.
Discussion: Our findings may help locate the most psychologically vulnerable adolescents during the epidemic and foster better implementation of targeted intervention.

IMPLICATIONS AND CONTRIBUTION

Using longitudinal latent class analysis, this study uncovered the heterogeneous trajectories of mental symptoms among adolescents back to school after the COVID-19 containment and identified the predictive factors associated with the high severity trajectories. Findings revealed adolescents' disparate psychological vulnerabilities and corroborated the need for the implementation of targeted intervention during an epidemic.

Ever since its first outbreak in December 2019, the COVID-19 epidemic has resulted in overwhelming impacts on the lives on the earth. To control the epidemic, a series of intervention measures associated with severe disruptions to normal daily life, such as home confinement, social distancing, and school closure,

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have been implemented compulsorily [1,2]. The concerns about disease itself as well as the unexpected drastic changes of social circumstances gave rise to a prevailing surge in emotional distress and tremendously increased the risk of public mental illness. Numerous studies have reported a rapid growth in the prevalence of mental symptoms, such as anxiety, depression, and insomnia, among a variety of populations during the outbreak of COVID-19 [3–6].

Children and adolescents are highly psychologically vulnerable due to their immature development of emotion-coping strategies [7]. More attention should be paid to their issues of mental health amid a public health emergency. The COVID-19 epidemic affected the mental health of children and adolescents in many ways [8], one of which was school closure and reopening. Compared to forced quarantine, school closure along with class suspension and home confinement, though much less disruptive, engendered far more ubiquitous effects on nearly the entire uninfected youth population. To this population, school closure inevitably brings about some negative consequences on both the physical and mental health: the associated social isolation, lack of necessary resources, high exposure to negative information via media and even drastic changes of basic lifestyle behaviors would easily aggravate the feelings of fear, loneliness, helpless, anxiety, panic, and depression [9–11]. Moreover, the transition from home schooling back to normal school life after school reopening makes the students confront formidable challenges to adapt quickly, which could further deteriorate their mental status.

Despite the anticipated unfavorable ramifications of school closure and reopening during the epidemic, limited studies had attempted to assess the actual psychological conditions and concrete symptoms of children and adolescents especially when the school resumed [12]. Surveys exploring the dynamics of mental health or the changes in mental symptoms of children and adolescents after school reopening are even scarcer. Furthermore, in-depth examination of different development patterns of mental symptoms is lacking for this population [13]. It is also unclear how the initiation and development of mental symptoms among children and adolescents is influenced by the epidemic and what groups are more mentally susceptible.

To fill the gaps in the above subjects, a longitudinal study was designed, which specifically explored a few puzzles about the mental health of children and adolescents after school reopening during the COVID-19 epidemic: Would the students experience a considerable level of mental symptoms, including depression, anxiety, and insomnia? Do diverging development courses of those mental symptoms exist among this population? How does the epidemic impact on the students' life and what is the associated psychological consequence? How to identify the groups with a higher risk of mental symptoms?

Methods

Study design and setting

This cohort study was conducted during the 2020 spring semester in an ordinary public junior high school located in the suburban area of a large city in the Jiangsu province, eastern part of China. Junior high school education in China spans 3 years: the first and second years are equivalent to grade 7 and grade 8 in the United States or year 8 and year 9 in England and Wales. The first year is also a transition phase from an elementary school to a high school. There were around 2,500 students enrolled in the school, and the student-to-teacher ratio was approximately 12. The schools in Jiangsu were closed for winter holiday on Jan 18th. Six days later, Jiangsu initiated the response to the COVID-19 emergency, extended the winter holiday, and implemented strict social distancing measures. When the schools reopened on April 7th and 13th for the first- and second-year students, respectively, the studied students had experienced a school closure of 79–85 days. During this period, other epidemic control measures included daily report on the travel information, lockdown of regions with infected cases, centralized or home-based medical management of COVID-19 cases and close contacts, epidemiology tracing and quarantine inspection for potentially infected persons, cleaning and disinfecting of public spaces, and restrictions on crowd-gathering activities in public areas.

All students attending their first and second years at the time of school reopening were eligible for surveillance ($n = 1411$, age range, 12–16 years). The surveys did not include the third-year students to refrain from interrupting their busy study schedule for the entrance examination of senior high school. Written consent was obtained from both the participating adolescents and their parents or guardians. The questionnaires were distributed and collected by the head teachers in the classroom. The researchers were provided with only the anonymous database after the survey data were released 2 years later for research purpose. No individual participant can be identified. This study is approved by the Ethics Review Committee of Xi'an Jiaotong-Liverpool University (No. ER-AOPHA-00100001211120221102184239) and conducted in accordance with the principles of the Helsinki Declaration. According to the local COVID-19 statistics, in the study region there was no COVID-19 outbreak during the period of surveillance (April–July 2020).

Using this database, a longitudinal study was designed, in which the first survey was set as the baseline and four subsequent surveys followed, each around 2–3 weeks apart, an interval adopted by abundant longitudinal studies of psychological symptoms with similar tools [14,15]. The baseline data were collected between April 22nd and April 28th, on average around 2 weeks after school reopening (mean $\pm$ standard deviation, 13.8 $\pm$ 2.5 days). The time intervals between the five surveys were 14.4 $\pm$ 2.2, 21.0 $\pm$ 3.3, 20.0 $\pm$ 3.8, and 12.6 $\pm$ 3.0 days, respectively. The last follow-up was arranged between June 30th and July 8th, while the semester ended on July 10th. The mean total observational duration was 68.4 $\pm$ 2.6 days. A total of 1,393 (98.7%) students participated in the cohort at the initial baseline, while 1,347 (95.5%), 1,340 (95.0%), 1,255 (88.9%), and 1,201 (85.1%) participants responded again in the second, third, fourth, and fifth follow-ups, respectively.

Questionnaire and variable

A paper-based questionnaire composed of three assessment tools, i.e. the nine-item Patient Health Questionnaire (PHQ-9), seven-item Generalized Anxiety Disorder scale (GAD-7), and seven-item Insomnia Severity Index (ISI), was used at each data collection point to evaluate the prevalence of depression, anxiety, and insomnia, respectively. The self-administered PHQ-9 contains nine items scored by a four-point Likert scale ranging from 0 (not at all) to 3 (nearly every day). The severity of depression symptoms is determined by the summed score of all items: normal (0–4), mild (5–9), moderate (10–14), moderate severe (15–19), and severe (20–27) [16]. GAD-7 is a self-reported scale for anxiety symptoms with seven items scored from 0 (not at all) to 3 (nearly every day). The total score of GAD-7 was categorized at the cutoff points of 5, 10, and 15, which represented increasing severity of anxiety (normal [0–4], mild [5–9], moderate [10–14], and severe [15–21]) [17]. The ISI is composed of seven items rated on a five-point Likert scale, ranging from 0 (not at all) to 4 (extremely). The total score was categorized in accord with increasing severity of insomnia: normal (0–7), subthreshold (8–14), moderate (15–21), and severe (22–28) [18]. The validity and reliability of the Chinese versions of PHQ-9, GAD-7, and ISI with the above cutoff points among Chinese adolescents have been proven by previous research [19–21]. The actual scores of PHQ-9, GAD-7, and ISI were used mainly for descriptive statistics, whereas the ordinal categorical data were used in cluster analysis.

Questions pertaining to sociodemographic factors and perceived impacts of COVID-19 outbreak on life were included in the baseline questionnaire. The perceived impacts of COVID-19 epidemic were investigated through five questions that referred to different aspects, including daily life, study practice, family income, family health, and family relationship. We assumed that the epidemic would only bring negative effects on the investigated life aspects, and thus, the five questions were scored by a five-point Likert scale based on the self-reported severity of negative epidemic impacts (0 = none, 1 = mild, 2 = moderate, 3 = severe, and 4 = extremely severe) (Table S1).

Statistical analysis

Descriptive statistics were firstly summarized for all the studied variables. Longitudinal latent class analysis (LLCA) was used to classify the participants into trajectory clusters according to their self-reported severity categories of depression, anxiety, and insomnia symptoms at each data collection point. Each participant was allocated to one cluster corresponding to depression, anxiety, or insomnia. In LLCA, the severity categories of depression, anxiety, and insomnia symptoms were treated as an ordinal variable, starting from 1 (as "normal"). LLCA parameters including the L^2 statistics, percentage reduction in L^2 from the null hypothesis (i.e. the model containing one cluster), Bayes information criterion, and consistent Akaike's information criterion were assessed to determine the optimal LLCA model (i.e. the optimal number of clusters). The analysis was performed on Latent GOLD (version 4.5; Statistical Innovations, Arlington, MA), which used the estimation-maximization and Newton-Raphson algorithms to estimate the model parameters [22]. One thousand different random starting values which include 100 iterations for each run were used. Lower Bayes information criterion and consistent Akaike's information criterion suggest a better model fit, and the optimal model is defined as a model with the smallest information criterion values. If these two information criterion values provide different suggestions, a model with a fewer number of clusters is selected. Participants were allocated to clusters based on their posterior probabilities of belonging to each cluster. An LLCA model can be considered to demonstrate clear allocation of participants to clusters when the derived mean posterior probability in each cluster is larger than 0.70 [23].

To determine factors predictive of trajectories of studied mental symptoms, a stepped process based on an approach we have used previously was employed [24]. First, univariate statistical analyses, such as analysis of variance and chi-squared tests where appropriate, were applied to reveal the crude association between each single risk factor and the clusters of each mental symptom. Multinomial logistic regression models were then used to assess the independent associations of the significant factors from the univariate analyses, with all variables entered simultaneously.

The analysis presented in the current report was based on the students participating in the baseline data collection (n = 1,393). We analyzed the original data according to the survey questions, unless additional data processing was mentioned elsewhere. The "missing values option" was activated in the Latent GOLD (Statistical Innovations), allowing for the inclusion of samples containing missing data. Including samples with missing values causes the mean to be inputted and handled directly in the likelihood function [22]. Sensitivity analysis including participants only with complete data (i.e. the "complete-case" approach) was also conducted. All the statistical analyses were performed with STATA (version 15; StataCorp LLC, College Station, TX), except for LLCA. Result visualization was done in GraphPad Prism 9 (GraphPad Software, Boston, MA). A p value <.05, two tailed, was set as the threshold for statistical significance.

Results

Characteristics of participants

The characteristics of all the participants (n = 1,393) at the baseline are summarized in Table 1. The mean age was 13.8 years; 53.3% were males and 52.3% were in their first year of school.

Table 1
Characteristics of all participants (n = 1,393) at the baseline

Variable	Number (%) or mean (standard deviation)
Age, years	13.83 (0.82)
Male	743 (53.3)
First year	729 (52.3)
Han Chinese	1,377 (98.9)
Place of birth[b]	
Jiangsu province	761 (56.1)
Outside of Jiangsu	596 (43.9)
Unknown	36
Body height, m[a]	1.64 (0.08)
Body weight, kg[a]	52.37 (10.79)
Body mass index, kg/m^2[a]	19.49 (3.52)
Having sibling(s)[b]	
Yes	956 (73.9)
No	338 (26.1)
Unknown	99
Having chronic condition(s), participant[b]	
Yes	12 (0.9)
No	1,326 (99.1)
Unknown	55
Having chronic condition(s), parent(s)[b]	
Yes	46 (3.5)
No	1,281 (96.5)
Unknown	66
Academic performance in the last semester, final average mark[b]	
≥90	143 (10.8)
≥80 & <90	365 (27.6)
≥70 & <80	398 (30.1)
<70	418 (31.6)
Unknown	69
Interval between the start of semester and study baseline, days	13.82 (2.47)

[a] Body height is available for 1,376 (98.8%) participants, body weight for 1,367 (98.1%) participants, and body mass index for 1,364 (97.9%) participants.
[b] Percentage excluding missing data (the "unknown" category).

Perceived impacts of COVID-19

The majority of the respondents reported that they had been influenced by the COVID-19 outbreak with varying extents (from "mild" to "extremely severe") on daily life (66.0%), study practice (69.3%), and family income (63.0%) (Table 2). Study practice accounted for the most concerns from the surveyed students: the percentages perceiving the impact on this aspect as "moderate," "severe," or "extremely severe" were all higher than those for the other aspects.

Prevalence of mental symptoms

The prevalence of possible depression among the respondents was 36.2% (mild, 20.2%; moderate, 8.2%; moderately severe, 4.8%; and severe, 3.0%) at the baseline and constantly falling at the subsequent follow-ups (Table 3). Analysis for the trend on the means of PHQ-9 scores over time demonstrated a highly significant decreasing curve (from 4.43 at the baseline to 2.71 at the fourth follow-up, p trend $< .001$). Similar trends were also observed in the symptom prevalence of anxiety and insomnia. Furthermore, over the study period, depression symptoms were always the most prevalent among the study participants.

The relationship between the perceived impacts of COVID-19 at the baseline and severity of mental symptoms over the study period demonstrated positive correlations (Table S2). In general, these correlations became weaker over the time, and some of them did not remain statistically significant at the last follow-up.

Trajectory analysis

The optimal models of the trajectories of depression, anxiety, and insomnia symptoms over the study period were determined sequentially among all the participants enrolled in this study. Statistical assessment of determining the optimal models is shown in Table S3. All the mean posterior probabilities for the assigned clusters were larger than 0.70, suggesting that the clusters were well distinguished from each other and the participants were clearly allocated to their assigned cluster (Table S4).

For the depression trajectory, participants were allocated into five distinctive clusters: 882 (63.3%) in the "resistance" cluster (no or very mild and stable symptoms, mean PHQ-9 scores: 1.5 at the baseline and 0.4 at the last follow-up), 226 (16.2%) in the "low symptom" cluster (low or moderate and stable symptoms, 7.6 and 6.8), 140 (10.1%) in the "recovery" cluster (initially moderate symptoms followed by a gradual decrease, 11.0 and 1.1), 104 (7.5%) in the "chronic dysfunction" cluster (severe and stable symptoms, 15.1 and 14.6), and 41 (2.9%) in the "delayed dysfunction" cluster (initially no or very mild symptoms followed by a sharp increase, 2.9 and 8.4) (Figure 1A). The "chronic dysfunction" cluster contained 70.2% first-year students, 56.7% females, and 64.3% of Jiangsu birth origin, all higher than those in other clusters (Table S5).

For the anxiety trajectory, five clusters with similar characteristics were found, but the proportions of participants distributed among them were slightly different: 926 (66.5%) in the "resistance" cluster (mean GAD-7 scores: 1.0 at the baseline and 0.2 at the last follow-up), 181 (13%) in the "low symptom" cluster (8.9 and 6.7), 134 (9.6%) in the "delayed dysfunction" cluster (2.3 and 4.8), 95 (6.8%) in the "recovery" cluster (10.0 and 0.7), and 57 (4.1%) in the "chronic dysfunction" cluster (12.8 and 14.9) (Figure 1B). The "Chronic dysfunction" cluster, which had the highest mean score of GAD-7, consisted of 61.4% females and 68.4% first-year students, all higher than the other four clusters (Table S6).

Regarding the insomnia trajectory, of which three clusters were determined, the largest cluster ("resistance," mean ISI scores: 2.4 at the baseline and 1.0 at the last follow-up) contained 1,063 (76.3%) participants, followed by the "low symptom" (with mean ISI scores: 8.9 and 6.8) and "chronic dysfunction" (with mean ISI scores: 13.8 and 14.7) clusters (251 [18.0%], and 79 [5.7%], respectively, Figure 1C). Among all the three clusters, the "chronic dysfunction" had the largest proportion of first-year students (69.6%) as well as the highest mean body mass index (BMI) (20.8) (Table S7).

With all the three symptoms considered simultaneously, 9.9% of participants (138/1,393) were associated with at least one of the "chronic dysfunction" clusters of the three studied symptoms, 7.0% (98/1,393) with at least two, and 2.2% (30/1,393) with all the three (Table S8). Participants allocated into the "resistance" clusters for all the three symptoms were 57.1% (795/1,393; Table S8).

Sensitivity analysis, which repeated the same analyses with only the participants who provided no missing value on PHQ-9, GAD-7, or ISI data at each data collection point, generated consistent trajectory patterns (Tables S9 and S10 and Figure S1).

Predictive factors

The baseline variables stratified by the clusters derived from LLCA were summarized for depression, anxiety, and insomnia, respectively (Tables S5–S7). The perceived negative COVID-19 impacts (with all the five components) were the greatest in the "chronic dysfunction" cluster (Table S8–S10).

Table 2
Perceived impacts of COVID-19 outbreak on five life aspects of all participants (n = 1,393) at the baseline

Aspect	Impact category, n (%[a])					Missing data, n
	None	Mild	Moderate	Severe	Extremely severe	
Daily life	473 (34.0)	564 (40.6)	287 (20.6)	34 (2.4)	33 (2.4)	2
Study practice	426 (30.7)	417 (30.0)	337 (24.2)	127 (9.1)	83 (6.0)	3
Family income	514 (37.0)	469 (33.8)	314 (22.6)	60 (4.3)	32 (2.3)	4
Family health	1,131 (81.4)	165 (11.9)	70 (5.0)	8 (0.6)	15 (1.1)	4
Family relationship	1,125 (80.8)	171 (12.3)	63 (4.5)	16 (1.2)	18 (1.3)	0

[a] Percentage excluding missing data.

Table 3
Symptoms of depression, anxiety, and insomnia of all participants (n = 1,393) over the study period

	Baseline	First follow-up	Second follow-up	Third follow-up	Fourth follow-up
PHQ-9					
Category (depression symptom), n (%)[a]					
Normal (0–4)	873 (63.9)	949 (70.9)	978 (73.4)	970 (77.5)	940 (78.3)
Mild (5–9)	276 (20.2)	217 (16.2)	212 (15.9)	144 (11.5)	156 (13.0)
Moderate (10–14)	112 (8.2)	95 (7.1)	64 (4.8)	61 (4.9)	40 (3.3)
Moderately severe (15–19)	65 (4.8)	44 (3.3)	47 (3.5)	41 (3.3)	33 (2.8)
Severe (20–27)	41 (3.0)	34 (2.5)	32 (2.4)	35 (2.8)	32 (2.7)
Missing data	26	54	60	142	192
Original score					
Median (IQR)	2 (0–7)	1 (0–6)	0 (0–5)	0 (0–4)	0 (0–3)
Mean (SD)	4.43 (5.75)[b]	3.64 (5.42)[b]	3.29 (5.51)[b]	3.01 (5.85)[b]	2.71 (5.50)[b]
GAD-7					
Category (anxiety symptom), n (%)[a]					
Normal (0–4)	990 (71.7)	1,027 (76.6)	1,016 (77.0)	982 (79.0)	975 (81.2)
Mild (5–9)	238 (17.2)	182 (13.6)	194 (14.7)	148 (11.9)	145 (12.1)
Moderate (10–14)	93 (6.7)	87 (6.5)	60 (4.6)	61 (4.9)	43 (3.6)
Severe (15–21)	60 (4.3)	45 (3.4)	50 (3.8)	52 (4.2)	38 (3.2)
Missing data	12	52	73	150	192
Original score					
Median (IQR)	0 (0–5)	0 (0–4)	0 (0–4)	0 (0–3)	0 (0–2)
Mean (SD)	3.25 (4.88)[b]	2.80 (4.62)[b]	2.56 (4.64)[b]	2.45 (4.84)[b]	2.12 (4.41)[b]
ISI					
Category (insomnia symptom), n (%)[a]					
Normal (0–7)	1,095 (79.8)	1,063 (79.2)	1,094 (82.3)	1,066 (85.1)	1,039 (86.5)
Subthreshold (8–14)	220 (16.0)	208 (15.5)	166 (12.5)	126 (10.1)	105 (8.7)
Moderate (15–21)	42 (3.1)	56 (4.2)	49 (3.7)	31 (2.5)	38 (3.2)
Severe (22–28)	16 (1.2)	15 (1.1)	21 (1.6)	30 (2.4)	19 (1.6)
Missing data	20	51	63	140	192
Original score					
Median (IQR)	3 (0–7)	2 (0–6)	1 (0–6)	0 (0–5)	0 (0–4)
Mean (SD)	4.25 (5.02)[b]	4.02 (5.25)[b]	3.59 (5.40)[b]	3.30 (5.75)[b]	2.84 (5.29)[b]

PHQ-9 = Patient Health Questionnaire nine-item scale; GAD-7 = Generalized Anxiety Disorder seven-item scale; ISI = Insomnia Severity Index; IQR = interquartile range; SD = standard deviation.
[a] Percentage excluding missing data.
[b] One-way analysis of variance trend analysis for the means over time, p trend < .001.

The multinomial regression model quantified the effects of risk factors significantly associated with the trajectory of more severe depression symptom, with the adjustment for other factors. These included perception of greater negative impacts of COVID-19 epidemic on study practice (relative risk ratio [RRR] 1.61 one unit score increase, 95% confidence interval [CI] 1.31–1.98; the "chronic dysfunction" cluster vs. the "resistance" cluster), on family income (RRR 1.39 one unit score increase, 95% CI 1.09–1.77), and on family relationship (RRR 1.96 one unit score increase, 95% CI 1.47–2.60). Also, being female (RRR 2.30, 95% CI 1.47–3.60), being in the first year of school (RRR 2.38, 95% CI 1.49–3.82) and having a place of birth within Jiangsu (RRR 1.77, 95% CI 1.11–2.83) were identified as risk factors (Table 4).

Significant risk factors of the trajectory with higher severity of anxiety included perception of greater negative impacts of COVID-19 epidemic on study practice (RRR 1.52 one unit score increase, 95% CI 1.16–2.00; the "chronic dysfunction" cluster vs. the "resistance" cluster), on family income (RRR 1.57 one unit score increase, 95% CI 1.16–2.14), and on family relationship (RRR 1.87 one unit score increase, 95% CI 1.34–2.62). Similarly, being female (RRR 3.09, 95% CI 1.70–5.63) and being in the first year of school were also found to be risk factors (RRR 2.17, 95% CI 1.18–3.98) (Table 4).

As to insomnia, the factors predictive of high severity trajectory were greater perceived negative impacts of COVID-19 epidemic on study practice (RRR 1.61 one unit score increase, 95% CI 1.28–2.01; the "Chronic dysfunction" cluster vs. the "resistance" cluster), on family relationship (RRR 1.55 one unit score increase, 95% CI 1.16–2.08), being in the first year of school (RRR 2.04, 95% CI 1.21–3.43), and higher BMI (RRR 1.09 one unit score increase, 95% CI 1.03–1.16) (Table 4).

Discussion

To the best of our knowledge, this is the first longitudinal study investigating the potential trajectories of depression, anxiety, and insomnia symptoms among the adolescents back to campus right after the COVID-19 containment. Distinct trajectories were identified for each symptom using advanced statistical modeling LLCA. Being female, being in the first year of school, and greater perceived negative impacts of COVID-19 on study practice, family income, and family relationship were identified as the common risk factors associated with the high severity trajectories.

Recent research highlighted the high level of mental health problems among children and adolescents in China during the COVID-19 outbreak [12,25]. In particular, evidence has been provided that the prevalence of depression symptoms among some children and adolescents significantly increased 2 weeks after school reopened, compared to that before the epidemic [26]. Similarly, the deterioration of sleep among adolescents after school reopening was suggested in a recent before-after study [27]. In our study, the information about the average psychological status of the studied students during the school closure or

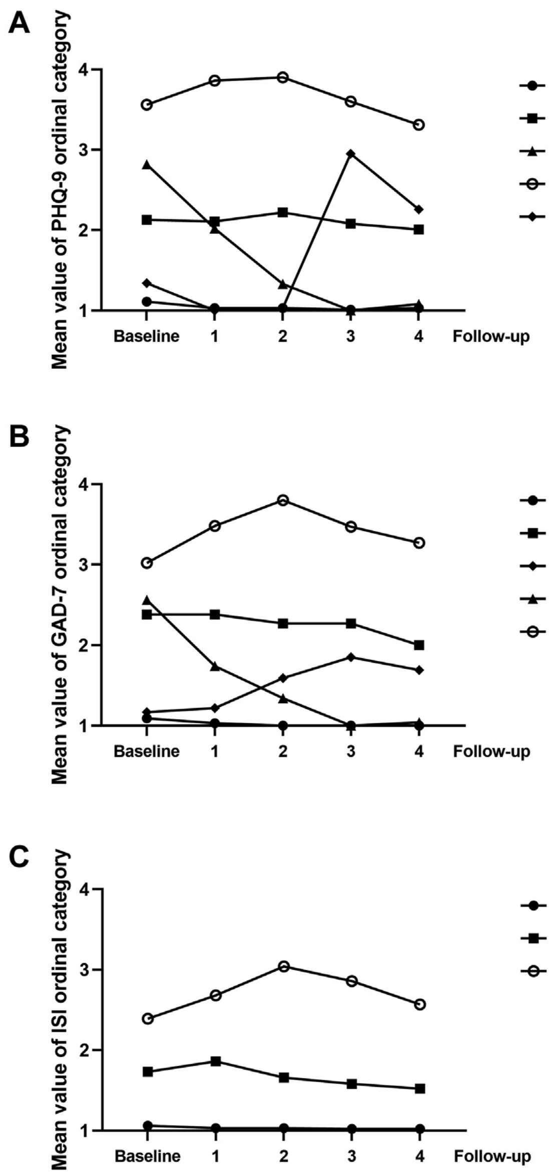

Figure 1. Trajectories of symptoms of depression (A), anxiety (B), and insomnia (C) over the study period (all participants, n = 1,393).

even before the COVID-19 outbreak was not available, but we did find that the prevalence of depression, anxiety, and insomnia symptoms was initially high but steadily decreasing after school resumed, which supposed a possible culmination of the mental symptoms around the time of school reopening. A similar trend of palliated depression and anxiety symptoms months after school reopening was also reported in another two-wave longitudinal study [28]. The transient occurrence of high-level mental symptoms after school reopening might be explained by two reasons: first, the uncertainty and panic caused by the COVID-19 outbreak imposed substantial stress to the students, which continued to accumulate in the course of school closure; and second, the sudden transition from confinement at home to normal school life brought about additional pressure for students to make prompt adjustment. Our study is also the first to assess the chronicity of these symptoms and suggests a considerable group of persistent symptoms as 9.9% of the participants reported chronic dysfunction for at least one of depression, anxiety, and insomnia.

According to our study, the surveyed students were allocated into five distinctive trajectory clusters for depression and anxiety and three for insomnia. It is noteworthy that these trajectories are frequently discovered in people's mode of psychological adaptation to severe stress [29,30]. A previous study among Chinese college students during the COVID-19 epidemic established five trajectories accordingly by presetting up a cutoff point for developing depression, anxiety, or insomnia and grouping participants based on their time-varying development of these symptoms [31,32]. Interestingly, the clustering results by LLCA in our study corroborated that model.

Our study also showed that students enrolled in the first grade, being female, having high BMI, and reporting greater

Table 4
Multinomial regression models for factors associated with symptom trajectories of depression, anxiety, and insomnia (n = 1,393)

Variables	Depression	Anxiety	Insomnia
	RRR (95% CI)	RRR (95% CI)	RRR (95% CI)
Sociodemographics			
Female (vs. male)			
Resistance	1.00 (referent)	1.00 (referent)	-
Low symptom	1.91 (1.41, 2.59)	2.11 (1.51, 2.96)	-
Delayed dysfunction	0.99 (0.52, 1.89)	1.44 (0.99, 2.08)	-
Recovery	1.16 (0.80, 1.69)	1.40 (0.90, 2.17)	-
Chronic dysfunction	2.30 (1.47, 3.60)	3.09 (1.70, 5.63)	-
Academic year 1 (vs. 2)			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	1.26 (0.92, 1.71)	1.78 (1.26, 2.51)	1.20 (0.90, 1.61)
Delayed dysfunction	1.20 (0.62, 2.29)	0.97 (0.67, 1.41)	-
Recovery	1.19 (0.82, 1.74)	1.12 (0.72, 1.74)	-
Chronic dysfunction	2.38 (1.49, 3.82)	2.17 (1.18, 3.98)	2.04 (1.21, 3.43)
Within (vs. outside) Jiangsu			
Resistance	1.00 (referent)	-	-
Low symptom	0.97 (0.71, 1.32)	-	-
Delayed dysfunction	0.89 (0.46, 1.71)	-	-
Recovery	0.67 (0.46, 0.98)	-	-
Chronic dysfunction	1.77 (1.11, 2.83)	-	-
Body mass index			
Resistance	-	-	1.00 (referent)
Low symptom	-	-	1.03 (0.98, 1.07)
Delayed dysfunction	-	-	-
Recovery	-	-	-
Chronic dysfunction	-	-	1.09 (1.03, 1.16)
Perceived COVID-19 impacts			
Daily life			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	1.02 (0.83, 1.25)	1.10 (0.88, 1.36)	1.12 (0.93, 1.35)
Delayed dysfunction	0.93 (0.61, 1.44)	0.86 (0.66, 1.11)	-
Recovery	0.98 (0.77, 1.24)	1.14 (0.87, 1.50)	-
Chronic dysfunction	0.87 (0.66, 1.14)	0.99 (0.70, 1.40)	1.03 (0.78, 1.37)
Study practice			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	1.35 (1.16, 1.57)	1.41 (1.19, 1.66)	1.25 (1.09, 1.45)
Delayed dysfunction	1.36 (0.10, 1.86)	1.33 (1.10, 1.59)	-
Recovery	1.22 (1.01, 1.48)	1.28 (1.03, 1.60)	-
Chronic dysfunction	1.61 (1.31, 1.98)	1.52 (1.16, 2.00)	1.61 (1.28, 2.01)
Family income			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	1.20 (1.01, 1.44)	1.14 (0.94, 1.38)	1.10 (0.93, 1.30)
Delayed dysfunction	0.81 (0.53, 1.22)	0.99 (0.79, 1.24)	-
Recovery	1.29 (1.05, 1.59)	1.15 (0.90, 1.47)	-
Chronic dysfunction	1.39 (1.09, 1.77)	1.57 (1.16, 2.14)	1.26 (0.97, 1.64)
Family health			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	0.81 (0.61, 1.07)	0.93 (0.71, 1.23)	0.97 (0.77, 1.23)
Delayed dysfunction	0.90 (0.50, 1.62)	1.21 (0.88, 1.65)	-
Recovery	1.37 (1.05, 1.78)	1.47 (1.10, 1.96)	-
Chronic dysfunction	0.91 (0.66, 1.26)	0.79 (0.52, 1.21)	1.10 (0.80, 1.51)
Family relationship			
Resistance	1.00 (referent)	1.00 (referent)	1.00 (referent)
Low symptom	1.59 (1.24, 2.03)	1.56 (1.22, 1.99)	1.55 (1.26, 1.91)
Delayed dysfunction	1.71 (1.06, 2.75)	1.32 (0.98, 1.78)	-
Recovery	1.31 (1.00, 1.72)	1.30 (0.97, 1.75)	-
Chronic dysfunction	1.96 (1.47, 2.60)	1.87 (1.34, 2.62)	1.55 (1.16, 2.08)

RRR = relative risk ratio; CI = confidence interval.
Risk factors simultaneously entered were gender, grade, place of birth, and perceived negative COVID-19 impacts on daily life, study practice, family income, family health, and family relationship in the model for depression; were gender, grade, and perceived negative COVID-19 impacts on daily life, study practice, family income, family health, and family relationship in the model for anxiety; and were grade, body mass index, and perceived negative COVID-19 impacts on daily life, study practice, family income, family health, and family relationship in the model for insomnia.

negative COVID-19 impacts especially on study practice, family relationship, and/or family income were more likely to suffer from severe and enduring psychological symptoms after school reopening. Female gender has been identified in a large body of literature as a risk factor of mental health problems [33,34]. Also, the association between COVID-19–related distress, particularly study difficulties, loss of family income, and deteriorating parent-child relationship, and worsening psychological status among children and adolescents has been revealed by previous studies [35,36]. The risk factors identified in this study imply that to the female, first-grade, or high BMI students who might be more affected by COVID-19 in study practice, family relationship, or income, school closure as a countermeasure to COVID-19 might incur more disadvantageous psychological consequences, such as higher levels of depression, anxiety, or insomnia. On the other hand, it should be noticed that the associations between the perceived negative impacts of COVID-19 and the mental symptoms were increasingly attenuated over the study period. This suggests that when adolescents return to normal school life, the negative impacts of COVID-19 epidemic on mental health would diminish within a couple of months, a finding also proposed by some other studies [37].

Some limitations should be noted for this study. First, our study participants were from a single school located in the Jiangsu province of China, and thus, the representativeness of the study sample may be compromised. Caution should be exercised when generalizing our findings to other populations under different contexts, e.g. foreign countries. Second, all the psychological symptoms studied were measured by self-reports rather than clinical diagnostics. The interference caused by subjective judgment to reported results, therefore, cannot be avoided. Nevertheless, the questionnaires used in our study are all well-established and widely recognized tools in psychological research. Third, we only investigated the symptoms of depression, anxiety, and insomnia. Other possible psychosomatic and somatic symptoms were not considered. However, since depression, anxiety, and insomnia are the most prevalent psychological symptoms among children and adolescents, our study still contributes to a better understanding about the mental problems of adolescents during the epidemic. Fourth, the assessment of the perceived negative impacts of COVID-19 was based on the assumption that overall, the epidemic was unlikely to favorably influence the students' life, which had not been attested though. Fifth, despite the several predictive factors identified by this study, other factors which might also influence the patterns of psychological status were neglected, such as physical exercise, access to entertainment, social support [38], and preexisting mental illness. Moreover, the students with a high risk of mental illness might have been given some interventions before or during the study period. But such information is not available in the current study. Sixth, the symptom trajectories in this study are determined by LLCA; however, using other analytical approaches (e.g. latent growth curve analysis) may generate slightly different results. Lastly, the 2-year deferral of our data acquisition and analysis to some extent undermined the timeliness and relevance of our findings, but our study still provided considerable insights into the mental health issues of children and adolescents during a severe epidemic and encouraged the development of precise psychological interventions to prepare for future public health emergencies.

In conclusion, the observation of heterogeneous trajectories of mental symptoms among adolescents back to school after the COVID-19 restriction not only illuminated the different patterns of psychological response to stress among the youth but also has important implications for the psychological intervention and clinical resource allocation amid the epidemic. The disparate trajectories followed by adolescents suggest their varying psychological immunity against stressful events, differing time period of symptom manifestation, as well as differential levels of resilience. Prioritized and immediate aids should be provided to those in the most urgent need in order to enhance the effectiveness of psychological intervention [39]. The factors predicting high-risk trajectories could help locate the most vulnerable groups of adolescents at an early stage of the epidemic based on their characteristics.

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Supplementary Data

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